GLA UNIVERSITY

Department of Computer Engineering & Applications

[Session : 2026-27]

MINI PROJECT SYNOPSIS ON

“PAWCARE-Pet Adoption & Services Platform”

Submitted By:

Samriddhi Bansal (2415001397)

Monika Chanchal (2415000991)

Meenakshi (2415000967)

Gaurav (2415000593)

Kajal Saraswat (2415000746)

Submitted To:

Mr. Akash Kumar Choudhary

(Technical Trainer)

INDEX

Introduction

About the Project:

The growing interest in pet adoption and pet ownership has created a need for an organized digital platform that can simplify the process of finding adoptable pets and accessing essential pet-care services. Currently, pet information, adoption opportunities, and care services are often available through separate and unorganized sources, making it difficult for users to obtain complete information and manage their requirements efficiently.

PawCare is proposed as a centralized MERN-based (MongoDB, Express.js, React.js, Node.js) web application that integrates pet adoption and pet-care services into a single platform. It enables users to explore available pets, view detailed pet information, submit adoption applications, and book required pet-care services such as veterinary consultation, grooming, walking, sitting, training, vaccination, and boarding. Administrators are provided with an efficient system for managing pets, users, applications, services, and bookings.

Primary Reason to Choose This Project

At present, information about pets available for adoption and about different pet-care services is spread across multiple, unorganized platforms - social media pages, NGO listings, and phone calls - making it difficult for a user to find everything at one place. PawCare is chosen as a project to solve this real, relatable problem by providing a single, organized platform for pet adoption and pet-care service booking. The project is also well-suited for applying core MERN Stack concepts such as frontend and backend development, database design, REST APIs, authentication, and CRUD operations in a practical, real-world application.

The Main Objective of The Project

The main objective of PawCare is to simplify the pet adoption process and to provide pet owners reliable, bookable pet-care services on a single platform. Specifically, the project aims to:

- Provide a centralized platform to browse and search pets available for adoption.
- Allow users to submit, view, and track adoption applications.
- Enable users to discover and book pet-care services such as vet consultation, grooming, walking, sitting, training, and boarding.
- Provide administrators with tools to manage pets, users, applications, service providers, and bookings.

- Implement secure, role-based access for adopters, service providers, and administrators.

Project Planning

The following plan will be followed for the development of the PawCare – Pet Adoption & Services Platform :

Phase 1: Requirement Analysis:

The main requirement of this project is to develop a web-based platform that provides pet adoption and pet-care services at one place.

The system will mainly support two types of users:

- User
- Administrator

Functional Requirements :

The system must fulfill the following functional requirements:

- Allow users to register and login.
- Display pets available for adoption.
- Provide detailed information about each pet.
- Allow users to submit adoption applications.
- Allow users to view and track their adoption applications.
- Display different pet-care services.
- Allow users to book available pet-care services.
- Provide booking details and booking status.
- Allow administrators to manage pets, users, adoption applications, services, and bookings.
- Provide a user-friendly and responsive interface.

Phase 2: System Design and Development

In this phase, the overall structure and working flow of the PawCare platform will be planned.

- Identify the major modules of the system.

- Design the overall system workflow.
- Design the database structure using MongoDB.
- Plan the frontend pages and navigation.
- Design the communication between frontend, backend, and database.
- Prepare Level 0, Level 1, and Level 2 DFDs.
- Define user and administrator functionalities.

Module Description

1. Authentication & Navigation

This module provides secure user registration and login, along with easy navigation through the main sections of the PawCare website.

2. Pet Adoption Module

This module allows users to browse available pets, view their details, and submit adoption requests. It also allows users to list pets for adoption, subject to admin approval.

3. NGO / Shelter / Pet Business Module

This module allows NGOs, shelters, and eligible pet businesses to register on the platform and list pets for adoption after verification and approval by the admin.

4. Pet Care Services Module

This module helps pet owners find and book services such as veterinary care, grooming, training, and pet walking. Service providers can manage their services and bookings through their dashboard.

5. Dashboard Module

This module provides separate dashboards for users and administrators. Users can track their adoption requests, pet listings, and service bookings, while administrators can manage users, pets, providers, requests, and bookings.

6. Resources – Pet Guide Module

This module provides useful pet-care information and guidance for pet owners. It includes a pet recommendation quiz that helps first-time users identify suitable types of pets based on their lifestyle and needs.

7. Database Layer

This layer stores and manages important information such as user accounts, pet details, adoption requests, service providers, services, and bookings, ensuring that the platform can access and manage information efficiently.

Phase 3: Frontend Development

The frontend of the application will be developed using React.js. The main purpose of this phase is to create an interactive and easy-to-use interface.

- Home page
- User registration and login
- Pet listing page
- Pet details page
- Adoption application form
- Pet-care services page
- Service booking page
- User profile
- Adoption application status
- Booking history
- Admin dashboard

The interface will be designed to be simple and responsive so that users can easily navigate through the platform.

Phase 4: Backend Development

In this phase, the backend of the application will be developed using Node.js and Express.js .The model learns patterns from the training data and then predicts the class of new, unseen voice samples.

The backend will be responsible for:

- Handling user registration and login.
- Managing user authentication and authorization.
- Managing pet information.
- Handling adoption applications.
- Managing pet-care services.
- Processing service bookings.
- Managing admin operations.
- Connecting the frontend with the database.
- Providing rest APIs for different operations.

CRUD operations will be implemented wherever required for efficient management of application data.

Phase 5: Database Development and Integration

In this phase, MongoDB will be used for storing and managing the data of the application. The major data collections will include:

1. Users
Stores user information such as name, email, password, contact details, and role.

2. Pets
Stores information such as pet name, species, breed, age, gender, location, health status, description, and adoption status.

3. Adoption Applications
Stores user adoption requests, selected pet details, application date, and application status.

4. Services
Stores information about available pet-care services, their description, duration, price, and availability.

5. Bookings
Stores user booking details, selected service, date, time slot, and booking status.

The database will be connected with the backend using Mongoose.

Phase 6: Authentication and Feature Integration

In this phase, all the major components of the system will be connected and integrated.

- Integrate React frontend with backend APIs.
- Connect the backend with MongoDB.
- Implement user authentication.
- Implement role-based access for users and administrators.
- Integrate pet search and listing functionality.
- Connect adoption applications with the database.
- Integrate pet-care service booking.
- Connect admin management features.

- Verify that data is properly transferred between the frontend, backend, and database.

Phase 7: Testing

In this phase, the application will be tested to check whether all the major functionalities are working properly.

- Test user registration and login.
- Test pet adoption and service booking.
- Test admin functions and database operations.
- Identify and fix errors.
-

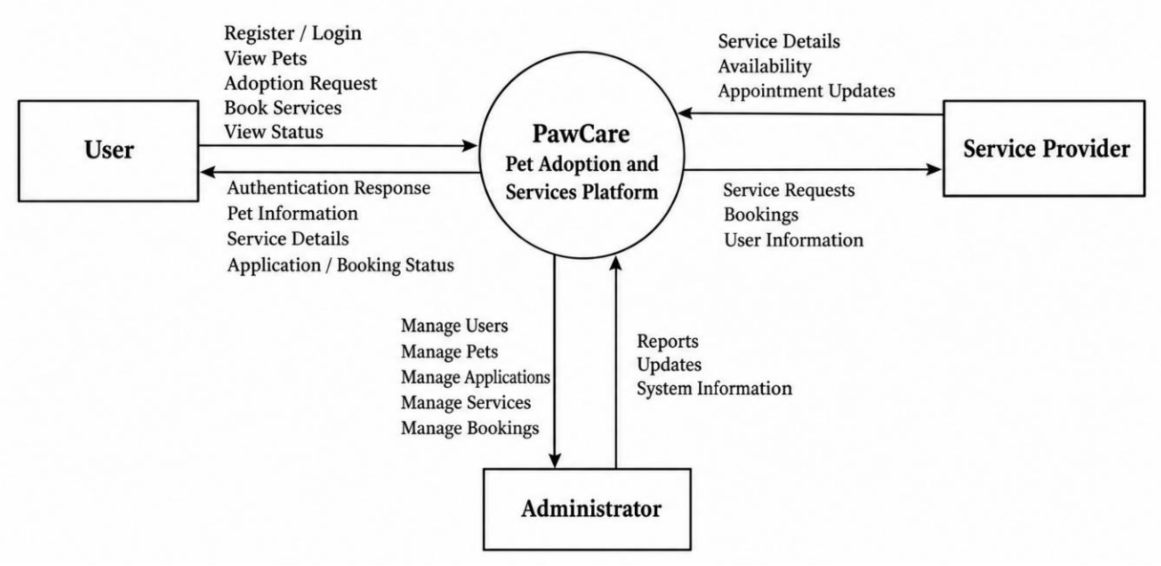
Phase 8: Deployment and Documentation

In the final phase, the completed project will be prepared for submission.

- Maintain the project using Git and GitHub.
- Prepare the final documentation and screenshots.
- Prepare the presentation and final project demonstration.

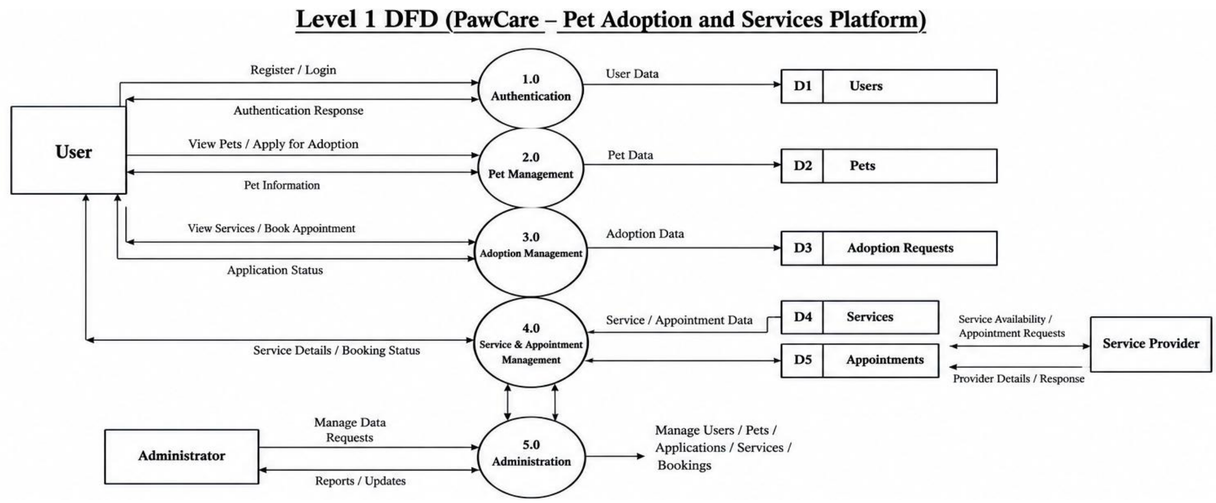
Data Flow Diagrams

0 Level DFD :



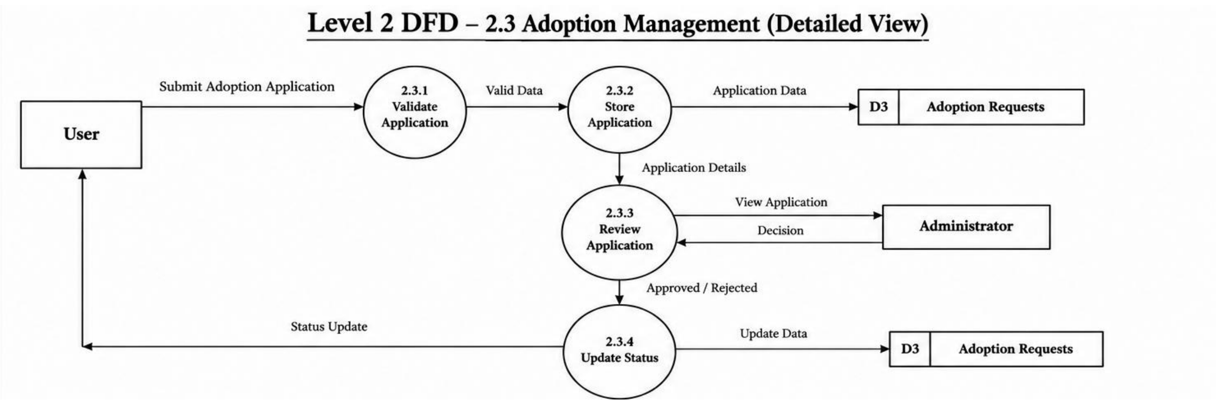
1 Level DFD :

The main system is divided into major processes:

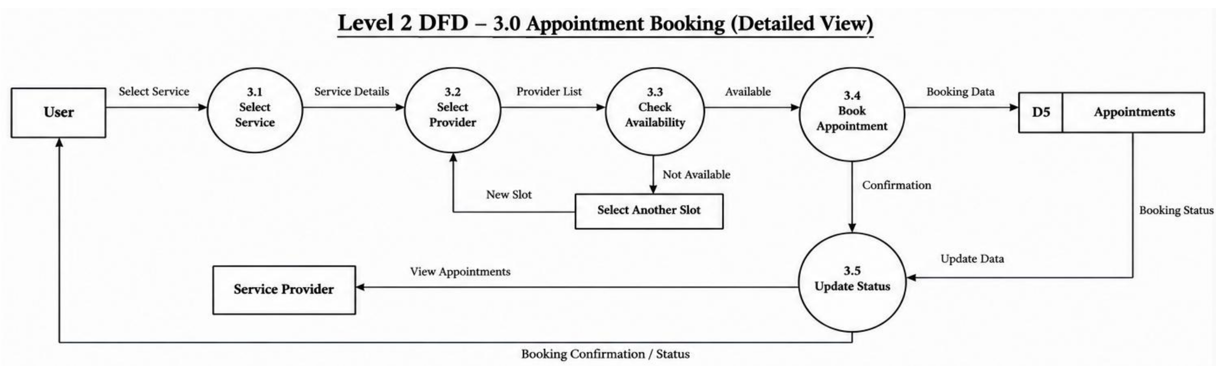


2 LEVEL DFD :

For the Adoption Management process:



For Appointment Management:



Tools Required

Software Requirement :

- Frontend: React.js, HTML, CSS, JavaScript
- Backend: Node.js, Express.js
- Database: MongoDB
- Database Tool: MongoDB Compass
- Code Editor: Visual Studio Code
- API Testing: Postman
- Version Control: Git and GitHub

Hardware Requirement :

- Processor: Intel Core i3 or equivalent
- RAM: 4 GB minimum
- Storage: 10 GB free space
- Keyboard and Mouse

References:

Existing / Related Platforms Studied :

- Juspets – India's first digital pet adoption platform, with discovery of nearby grooming, walking, boarding, training, vet, and NGO services – https://www.juspets.in/
- PetNimbus – Subscription-based pet adoption app with AI compatibility scoring, vaccination tracking, nutrition plans, and bookable vet/grooming/daycare services – https://www.petnimbus.com/
- PetBond – Pet-care super-app with a 24/7 AI health assistant, verified vet/groomer/hostel bookings, and an NGO-Vet-Foster-Sponsor adoption pipeline – https://www.petbond.in/
- VetEra – Pet-health-first platform for vet appointment booking, AI health monitoring, and 24/7 emergency vet assistance – https://www.vetera.in/
- Pawtners – Pet-services marketplace listing verified providers for grooming, boarding, walking, training, and veterinary care, with a pets-for-adoption section – https://www.pawtners.in/
- Zupaws – At-home pet-care platform offering vet visits, vaccination, diagnostics, grooming, and pet adoption/purchase – https://zupaws.in/

Sign of Project Guide: ____________________